

**EXAM 7 – SPRING 2014**

20. (1.5 points)

a. (1 point)

Describe an example where two lines of insurance that are typically uncorrelated become correlated during a stressed or extreme event.

b. (0.5 point)

Describe a method to account for increasing correlation in the tails of the loss distributions of typically uncorrelated risks.

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## SAMPLE ANSWERS AND EXAMINER'S REPORT

| QUESTION 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                      |
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| <b>TOTAL POINT VALUE: 1.5</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>LEARNING OBJECTIVE:</b> C2: Describe the use of enterprise-wide risk modeling and aggregation techniques; and C3: Evaluate and select appropriate models to handle diverse risks, including stochastic approaches |
| SAMPLE ANSWERS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                      |
| <b>Part a: 1 point</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                      |
| <p><i>Sample Answer 1</i></p> <p>Auto and homeowners aren't typically correlated. If someone gets into an auto accident, the majority of the time it's with another car, not a house. However, during a hurricane, there will be many homes and cars destroyed, causing these two lines to be correlated.</p> <p><i>Sample Answer 2</i></p> <p>Mortgage insurance in Canada and Workers' Compensation in U.S. are uncorrelated in normal circumstances because the two lines are in different industries, claims are triggered by different events, and not in the same geographical area, facing different policyholders with minimal interaction.</p> <p>In the event of global economic recession, the unemployment rate goes up and hence mortgage defaults as well as increased claims on Workers' Compensation from workers avoiding layoffs and to get income.</p> |                                                                                                                                                                                                                      |
| <b>Part b: 0.5 point</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                      |
| <p><i>Sample Answer 1</i></p> <p>We could use a copula to join two marginal distributions into a joint distribution. Based on our selection of the copula, we can choose how we want the dependencies to vary across the entire distribution.</p> <p><i>Sample Answer 2</i></p> <p>Find a copula, such as the HRT, to combine the distribution of commercial property risks and the distribution of Workers' Compensation risks. This copula will account for high correlation in the tails but will not show much correlation in the middle of the distribution.</p>                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                      |
| EXAMINER'S REPORT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                      |
| Generally candidates performed well on this question. Most candidates were able to answer both parts well.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                      |
| <b>Part a</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                      |
| <p>Most candidates were able to identify two uncorrelated lines of insurance and describe an event that would lead to correlation of the two lines of insurance.</p> <p>Candidates generally lost points when they did not describe how two lines are ordinarily uncorrelated. Some candidates lost points by not fully describing how an event can cause an increase in correlation. A minority of candidates also lost points when they wrote "catastrophe" or</p>                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                      |

## SAMPLE ANSWERS AND EXAMINER'S REPORT

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| "extreme event", instead of specifying the event (e.g., earthquake).                                                                                                          |
| <b>Part b</b>                                                                                                                                                                 |
| Overwhelmingly, candidates responded with "copula". Candidates lost points on this part either by not fully describing a copula or by giving a wrong description of a copula. |